Tectonophysics, 139 (1987) 1–20
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

Palaeomagnetism, palaeogeography and the multicomponent analysis of Middle and Upper Cambrian rocks of the Barrandian in the Bohemian Massif

MIROSLAV KRS [1], MARTA KRSOVÁ [1], PETR PRUNER [1], RICHARD CHVOJKA [1] and VLADIMÍR HAVLÍČEK [2]

[1] *Geofyzika, N.C., Brno, Prague Division, Geologická 2, 152 00 Prague 5—Barrandov (Czechoslovakia)*
[2] *Geological Survey, Malostranské nám. 19, 118 21 Prague 1—Malá Strana (Czechoslovakia)*

(Received February 2, 1986; revised version accepted July 15, 1986)

Abstract

Krs, M., Krsová, M., Pruner, P., Chvojka, R. and Havliček, V., 1987. Palaeomagnetism, palaeogeography and the multicomponent analysis of Middle and Upper Cambrian rocks of the Barrandian in the Bohemian Massif. In: D.V. Kent and M. Krs (Editors), Laurasian Paleomagnetism and Tectonics. Tectonophysics, 139: 1–20.

With a view to the importance of the Barrandian in the Bohemian Massif for global palaeogeographic reconstructions, considerable attention has been devoted to palaeomagnetic stability tests and multicomponent analysis of rocks of the pre-Variscan formations, using substantially improved laboratory techniques. This study is devoted to samples of Upper Cambrian andesite and Middle Cambrian sediments. Alternating and thermal fields were used for magnetic cleaning. A thermal demagnetizer fitted with a newly developed inductor of high sensitivity, the ROCOMA (rotating coil magnetometer), with a sensitivity of 0.1 to 0.2 nT, was used. The whole newly built and fully tested MAVACS system (magnetic vacuum control system) guaranteed a high degree of compensation of the magnetic field in a volume of 5 l, to a value not exceeding 2 nT. As numerous laboratory experiments have shown, the high magnetic vacuum was necessary to derive the palaeomagnetic directions of rocks of the older formations affected significantly by the Variscan orogenic phase. The position of the Upper Cambrian pole (28.6 °S, 131.2 °E) should be considered tentative because it was derived using samples from only three andesite outcrops. Two other andesite outcrops preserved Carboniferous or Permo-Carboniferous palaeomagnetic directions. The position of the Middle Cambrian pole (11.3 °S, 154.0 °E) was determined using samples of sediments (mainly greywackes and shales) from ten outcrops, some of which represent extensive rock cross-sections. By combining multicomponent analysis with fold tests, the mean direction of the pre-Variscan magnetization could be derived. The very low values of the moduli of the natural remanent magnetization of the Middle Cambrian sediments, and the considerable amount of magnetically soft components, required special care to be taken in the laboratory procedures. The palaeozoogeographic zones of the Barrandian are related to palaeogeographic latitudes of about 40 °S, derived from palaeomagnetic data. Cold water is characteristic for the Bohemian fauna, but arctic conditions can be ruled out due to the amount of primary red sediments, mostly of continental origin.

Introduction

This study is a continuation of the palaeomagnetic research into the Phanerozoic rocks of the Bohemian Massif (Czechoslovakia). Rocks of the Lower Permian (Autunian and Zechstein, collected in previous years from numerous outcrops in the Bohemian Massif) and Tertiary have yielded palaeomagnetic results corresponding to the palaeomagnetism of similar rocks of the North

0040-1951/87/$03.50 © 1987 Elsevier Science Publishers B.V.

European Platform. Rocks of Westphalian B to Stephanian age, of the "red bed" type, and tuffites collected from numerous surface and underground outcrops in four coal basins, have yielded palaeomagnetic directions with a larger scatter than the Lower Permian rocks. This scatter was interpreted in relation to the Asturian phase of Variscan folding and in relation to the block structure of the Bohemian Massif (a review of the basic palaeomagnetic data can be found in Krs (1968)

Devonian and older rocks presented a number of problems in the laboratory investigation of palaeomagnetic parameters. Devonian limestones frequently contain a high percentage of magnetically very soft components. Silurian, Ordovician and Cambrian rocks were often subject to phase changes of minerals in the course of the thermal treatment of specimens. Magnetization components which were apparently generated in the course of Variscan folding (see, e.g., Krs, 1978), were found in a number of pre-Variscan rock formations.

For the above reasons it was necessary to substantially improve the magnetic cleaning methods of Older Phanerozoic rocks, especially those which have multicomponent magnetizations. This is also the reason why multicomponent analysis of magnetization was applied. A newly developed instrument for thermal cleaning of rocks, the MAVACS (magnetic vacuum control system), was used to study the palaeomagnetic stability and for magnetic cleaning. This instrument in equipped with a highly sensitive inductor, the ROCOMA (rotating coil magnetometer).

To make the derived palaeomagnetic data as reliable as possible, rock samples of various geneses were subjected to palaeomagnetic study. Rocks of the Barrandian have been thoroughly studied biostratigraphically so that the palaeogeographical latitudes, derived from palaeomagnetic data, can be related to the palaeozoogeographical zones.

Outline of geology

The Lower Palaeozoic of Central Europe is part of the Hercynian mobile belt between the East European Platform and the former Gondwanaland; the tectonomagmatic process was active during the Upper Proterozoic and continued to the Lower Palaeozoic. A system of narrow linear basins, usually trending SW–NE or NW–SE (Havlíček, 1980), originated in Central Bohemia where, however, the Cambrian sediments overlie the folded and often metamorphosed Upper Proterozoic sequence with an angular unconformity. The Lower Cambrian deposits, over 2000 m thick and mostly of continental origin, are confined to a narrow intermontane depression of a graben-like nature, predisposed tectonically, the longitudinal axis of which strikes 40°–45°; Havlíček (1980) called this depression the Příbram–Jince Basin. Unlike other regions of Central Europe, the Cambrian rocks of Central Bohemia did not undergo regional metamorphism (contact metamorphism can be observed in the southwest of the Příbram–Jince Basin). The Cambrian sequence was slightly faulted and locally folded.

The Bohemian Lower Cambrian mainly consists of sandstones and conglomerates deposited in an environment of alluvial fans, alluvial plains and, probably, bays. The stratigraphy, lithology, weak volcanic activity and gradual deformation of the basin during the Lower Cambrian have been dealt with by Havlíček (1971) and Kukal (1971).

Five formations were assigned to the Lower Cambrian, namely the Žitec–Hluboš, Sádek, Holšiny–Hořice, Klouček–Čenkov and Chumava–Baština Formations, all lacking fossils except for the Holšiny–Hořice Formation which contains an aglaspid merostome *Kodymirus vagans*, most probably of limnic origin (Chlupáč and Havlíček, 1965).

In contrast to the Lower Cambrian, the Middle Cambrian shales and greywackes are mostly marine, as is evident from the richly diversified trilobite fauna in the Jince Formation. The earliest fossiliferous horizon corresponds to the middle/upper part of the Scandinavian *Paradoxides oelandicus* Zone, and the top of the Jince Formation is comparable with the middle part of the Scandinavian *Paradoxides paradoxissimus* Zone. The exact Lower–Middle Cambrian and Middle–Upper Cambrian boundaries are questionable because the trilobites, shelly fauna and microfossils have not been discovered yet in the supposed boundary beds.

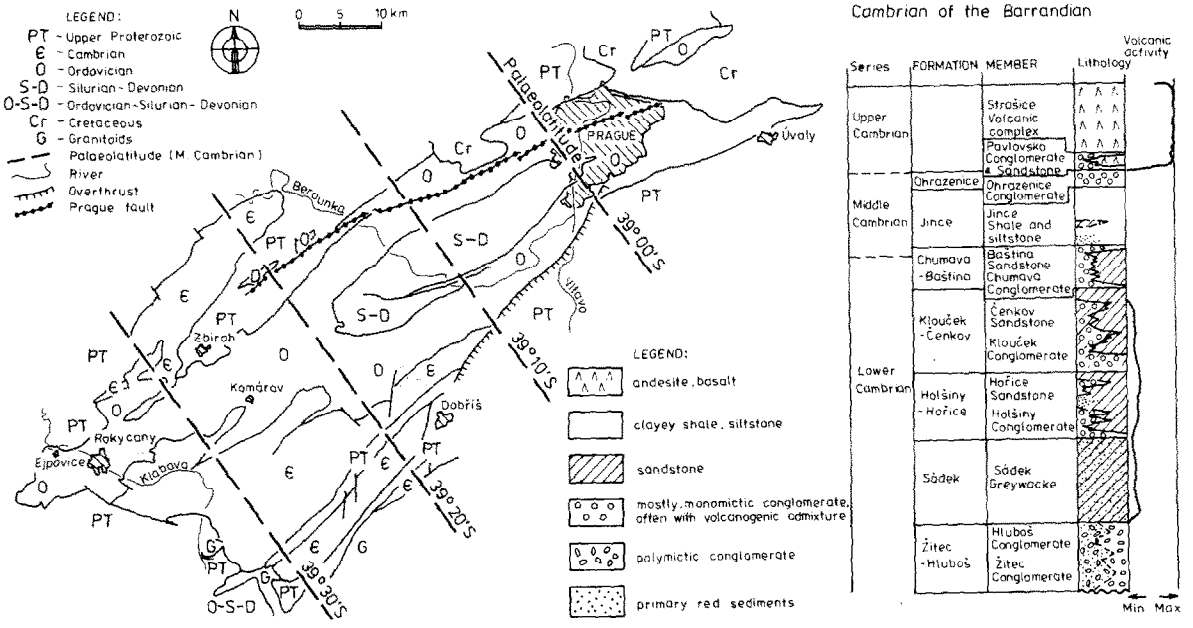


Fig. 1. Geological sketch of the Barrandian area, Bohemian Massif. Review of lithological units, Middle Cambrian palaeolatitudes calculated from the palaeomagnetic data presented in this study.

The shape of the Middle Cambrian basin differs markedly from that of the Lower Cambrian. According to Havlíček (1980), a "new marine way" originated, connecting the Příbram–Jince Basin with the Frankenwald area in Bavaria; at that time, marine transgression used the Skryje–Týřovice area as a strait, in which the Middle Cambrian sediments overlie the pre-Palaeozoic bedrock. The biostratigraphy, lithology and palaeogeography of the Middle Cambrian have been reported by Havlíček (1971) and Kukal (1971). Volcanic activity is absent.

In Central Bohemia, the Late Cambrian was a period of pre-Ordovician faulting and strong volcanic activity. At that time, two outstanding subparallel complexes arose, both utilizing the pre-Ordovician fault zones. The Křivoklát–Rokycany Complex, confined to the Skryje–Týřovice area, contains rocks of the andesite–rhyolite association, while the Strašice volcanic Complex, roughly bordering the Příbram–Jince Basin, contains rocks of the andesite–rhyolite–basalt association; the volcanic activity in both zones persisted into the Early Ordovician. According to Waldhausrová (1966, 1972), the Křivoklát–Rokycany and Strašice Complexes should be assigned to the post-Cadomian subsequent volcanism. Moreover, the fracturing of the bedrock was also recorded in the Příbram ore field where most of the pre-Ordovician fissures were filled with dykes of basalts, usually trending N–S.

Sampling for palaeomagnetic research

Upper Cambrian andesites as well as Middle Cambrian sediments, mostly shales and greywackes (see Table 1) were subjected to palaeomagnetic research. Hand samples of andesite were collected from the Skryje–Týřovice area, and samples of sediments were collected from the Skryje–Týřovice and Příbram–Jince areas (see Fig. 2). On the whole, 21 groups of rocks, corresponding to 21 rock outcrops with precise stratigraphic classifications, were sampled. The sediments were sampled by collecting each sample from a different layer; the same layer may, of course, have been sampled in another outcrop. Every sample is marked with an ordinal number and the letter A. If a number of specimens in the shape of a cube were prepared from a single sample in the laboratory, the specimens are marked A1, A2, etc. The collection of samples represents heterogeneous

TABLE 1

Review of oriented samples of Cambrian rocks collected for palaeomagnetic investigations in this study, Barrandian Area, Bohemian Massif, Czechoslovakia

Group no.	Sample nos.	Locality	Stratigraphy	Rock type
1	2971A-2988A	North of Týrovice	Upper Cambrian	andesite
2	2989A-3005A	50 m from previous outcrop towards Týřovice	Upper Cambrian	andesite
3	3006A-3018A	50 m from previous outcrop towards Týřovice	Upper Cambrian	andesite
4	3567A-3578A	Zaječov	Upper Cambrian	andesite
5	3583A-3588A	Jivina	Upper Cambrian	andesite
6	2957A-2970A	North of Týřovice, defilé along Berounka River	Middle Cambrian, Jince Formation, Vosník Member	greywacke, finegrained
7	3019A-3031A	Near Skryje, close to the bridge over Berounka River, entrance to Karásek's Gorge	Middle Cambrian, Jince Formation, Skryje Member	shale
8	3032A-3046A	A trilobite locality, 200 m from the cottage towards the hill	Middle Cambrian, Jince formation, Skryje Member	shale
9	3047A-3061A	Near Skryje, between the bridge and Skryje	Middle Cambrian, Jince Formation, Skryje Member	shale
10	3062A-3076A	300 m from previous outcrop towards Skryje, road bend	Middle Cambrian, Jince Formation, Skryje Member	shale
11	3077A-3092A	Buchava–Slap	Middle Cambrian, Jince Formation, Skryje Member	greywacke
12	3093A-3113A	30 m from previous outcrop towards Slap	Middle Cambrian, Jince Formation, Skryje Member	greywacke
13	3486A-3491A	Vinice, close to Jince	Middle Cambrian, basis of Jince Formation	greywacke
14	3492A-3497A	Along Litavka River, approx. 60 m NW from previous outcrop	Middle Cambrian, Jince Formation	greywacke, shale
15	3498A-3500A	Along Litavka River, 40 m NW from previous outcrop	Middle Cambrian, Jince Formation	greywacke, shale
16	3501A-3510A	Along Litavka River, 30 m N from previous outcrop	Middle Cambrian, Jince Formation	greywacke, shale
17	3511A-3515A	Along Litavka River, 30 m NW from previous outcrop	Middle Cambrian, Jince Formation	greywacke
18	3516A-3523A	Rejkovice, Hejdův dvůr	Middle Cambrian, Jince Formation	greywacke

TABLE 1 (continued)

Group no.	Sample nos.	Locality	Stratigraphy	Rock type
19	3524A-3530A	Rejkovice, 70 m from previous outcrop, towards the hill	Middle Cambrian, Jince Formation	greywacke
20	3531A-3538A	Rejkovice, 50 m from previous outcrop towards the hill	Middle Cambrian, Jince Formation	greywacke
21	3539A-3544A	Rejkovice, 40 m from previous outcrop against the stream	Middle Cambrian, Jince Formation	greywacke

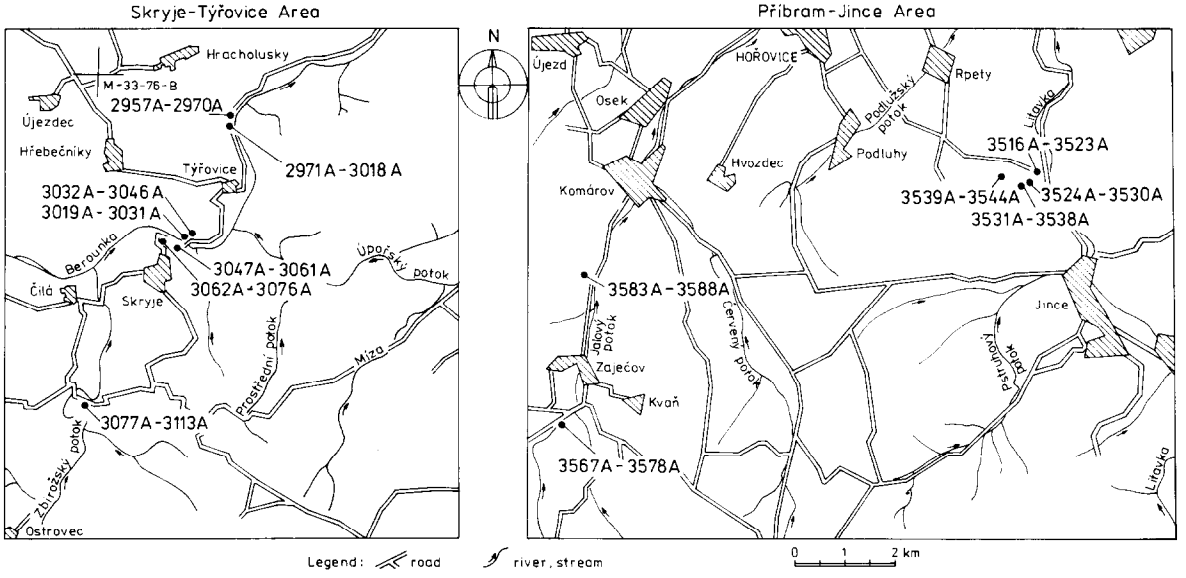


Fig. 2. Sketch maps showing the localities where oriented samples of Cambrian rocks were collected from the Skryje–Týřovice and Příbram–Jince areas of the Barrandian (see Table 1).

rock material, and the samples were intentionally collected from rock outcrops with a relatively simple tectonic evolution. Samples from quarries or more deeply eroded valleys were given preference in order to minimize the effects of weathering.

The basic magnetic properties of the studied rocks

Table 2 gives the basic magnetic properties of the studied Cambrian rocks, i.e. the arithmetic average ($\bar{x}$) of the modulus of natural remanent magnetic polarization (NRM - J_n), the volume magnetic susceptibility, κ, the values of the Q_n-coefficients (the Koenigsberger ratio) and the appropriate standard deviations (s). A JR-4 spinner magnetometer (Jelínek, 1966), and occasionally a LAM-24 astatic magnetometer (Krs, 1969), were used to measure the RM, and a KLY-2 kappa bridge (Jelínek, 1973) was used to measure the magnetic susceptibility.

The andesite samples are moderately magnetic rocks and the samples of the sediments are exceptionally magnetically weak. Some of the sediments are so magnetically weak that the whole cycle of thermal demagnetization of those samples could not be completed because of the limited sensitivity of the spinner magnetometers, which are among the most sensitive instruments of their kind. The

TABLE 2

Magnetic properties of Cambrian rocks, Barrandian area

Group no. of samples (locality)	Number of measured samples (n)	J_n (nT)		$\kappa \times 10^{-6}$ [SI]		Q_n	
		$\bar{x}$	s	$\bar{x}$	s	$\bar{x}$	s
1	51	180.13	±244.83	3465.5	±3006.9	0.980	±1.583
2	55	123.67	±167.47	10885.5	±6151.3	0.364	±0.825
3	38	196.53	±173.38	4207.8	±2610.8	0.916	±0.449
4	20	193.00	±37.40	17451.0	±3112.0	0.220	±0.030
5	11	101.00	±13.90	13689.0	±647.0	0.140	±0.02
6	49	0.185	±0.077	182.9	±42.3	0.021	±0.011
7	44	0.075	±0.031	320.9	±12.4	0.005	±0.006
8	48	0.190	±0.062	346.2	±15.8	0.011	±0.003
9	37	1.078	±2.879	329.4	±24.6	0.063	±0.182
10	55	0.270	±0.109	315.9	±34.3	0.020	±0.024
11	59	0.184	±0.062	317.1	±21.5	0.011	±0.004
12	81	0.179	±0.072	312.2	±15.5	0.011	±0.005
13	4	2.49	±0.09	165	±3.1	0.30	±0.02
14	4	5.20	±0.24	284	±12.7	0.37	±0.03
15	4	4.35	±0.16	274	±2.8	0.32	±0.01
16	4	1.84	±0.26	317	±7.6	0.11	±0.01
17	4	0.89	±0.12	287	±9.4	0.06	±0.01
18	4	2.75	±0.34	350	±9.5	0.16	±0.02
19	4	6.97	±0.72	421	±3.7	0.33	±0.03
20	4	4.24	±0.18	234	±2.3	0.36	±0.02
21	4	3.16	±0.42	378	±5.2	0.17	±0.02

components of the characteristic magnetization, derived by multicomponent analysis and used for the interpretation, displayed values of tens of pT in all sedimentary rocks. For this and other reasons, statistical methods were used to analyse the mean directions of the respective components of the RM.

Magnetic cleaning and multicomponent analysis of RM

Alternating field (AF) and thermal methods were used for magnetic cleaning and stepwise demagnetization of the specimens. AF demagnetization was used to a lesser extent because of its low efficiency; the Schonstedt GSD-1 instrument was employed for this purpose. The thermal tests were carried out exclusively with the newly developed thermal demagnetizer; the experimental data obtained with this instrument are described in this paper and all are indicative of its high efficiency. In developing the instrument, emphasis was put on being able to demagnetize at least 30 specimens at the same time, the furnace space being compensated to a value not exceeding 2 nT.

The thermal demagnetizer mentioned above was developed by Halousek, Pešina and Příhoda (1985). The non-magnetic furnace is located in a system of three Helmholtz coils which generate a triaxial homogeneous magnetic field which is necessary to compensate the Earth's field and to create a magnetic vacuum about 5 litres in volume with an accuracy of ±2 nT. An important part of the device is the newly developed highly sensitive rotating coil magnetometer (ROCOMA) with non-magnetic field sensors. Essentially, this involves an innovation along the lines of the Earth's inductor: the pick-up coil is placed in a rotor which is carried and rotated at a rate of 20000 rev/min by compressed air. The mean diameter of the coil is only 2 cm. The magnetometer consists of the air unit, the electronic unit and two sensors,

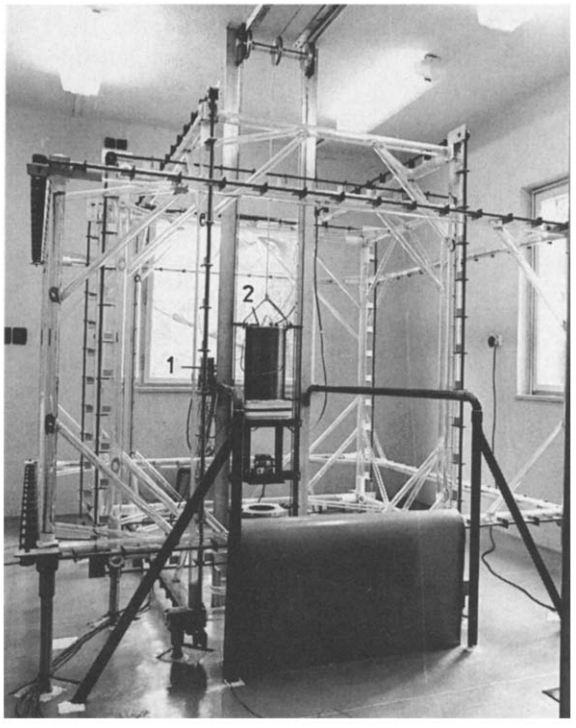


Fig. 3. Helmholtz induction coil system for automatic triaxial compensation of the Earth's magnetic field. The electric current flowing through the coils is continually controlled by two sensitive magnetometers (ROCOMA) to form a magnetic vacuum. *1*—sensor ROCOMA; *2*—non-magnetic furnace.

each of which enables two perpendicular components of the vector of the magnetic field intensity to be measured. The sensors guarantee detection of the zero magnetic field with an accuracy of ± 0.2 nT. The whole system, including the large stable Helmholtz coils, is known as MAVACS (magnetic vacuum control system). Figure 3 shows the system of Helmholtz coils with the furnace and sensors, and Fig. 4 shows the electronic control unit, the air unit and the sensor itself.

As we shall see later, the rocks of the localities being studied all display multicomponent RM. The directions of the individual magnetic components were determined using the procedure described by Kent et al. (1983). The calculations were made using the original unsmoothed values of the variance of the measurement errors. This variance displays systematic changes at many localities which depend on the temperature under which the specimens are demagnetized. A relatively large increase in the variance was observed at temperatures of 100°–150°C. This is apparently due to chemical changes in the specimens in the course of thermal demagnetization.

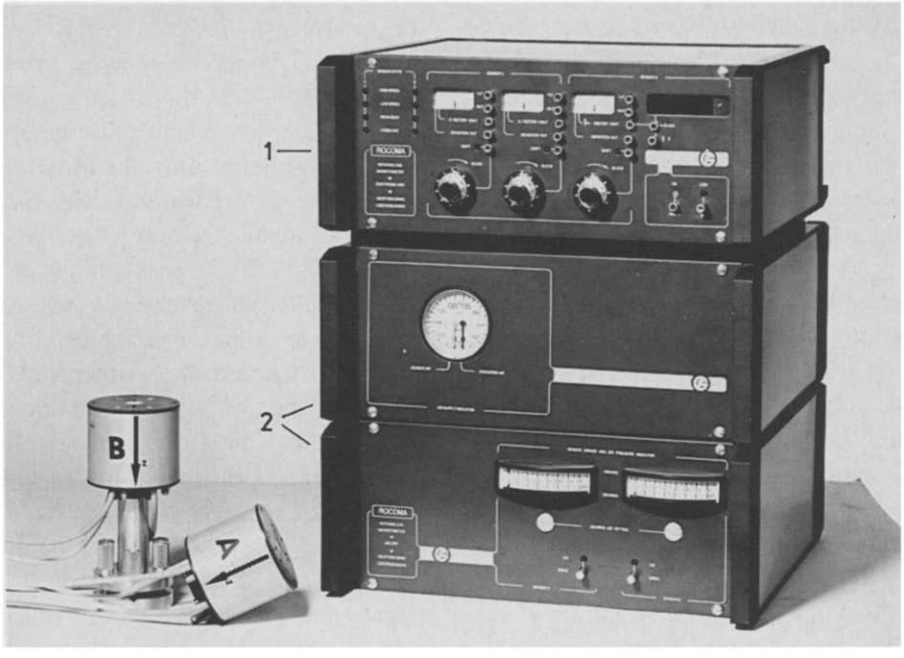


Fig. 4. Rotating coil magnetometer (ROCOMA). *1*—Electronic unit processing three signal channels corresponding to the magnetic intensity components. *2*—Air unit controlling the air supply to the sensor rotor drive and bearing. *A*, *B*—two sensors each sensing two magnetic intensity components: offset of the sensor is better than ± 0.2 nT. (Developed by J. Halousek, B. Pešina an K. Příhoda, Division of Geofyzika.)

The results of analysis of the thermal demagnetization data are illustrated in orthogonal projections; the linear segments found are represented by line sections. The end points of the line sections correspond to the orthogonal projections of the end points of the linear segments onto the straight lines passing through the weighted means of the points of the linear segments. The directional vectors of the straight lines correspond to the directions of the individual magnetic components. The projections of the end points of the linear segments were determined independently in the horizontal and vertical planes. The graphs also show the linear segments determined with an angular error larger than the recommended value of 20°, quoted by the authors mentioned above, however, making the assumption that these magnetic components could be clearly distinguished in most of the specimens from the locality in question. Some of the linear segments illustrated may, or course, be a combination of two or more magnetic components.

The thermal demagnetization of the other specimens was carried out on the basis of the analysis of pilot samples. The number of demagnetization steps and the temperature values were chosen so that the magnetic component being sought would be reflected in four steps, provided that a component with the most stable magnetization was not involved, or in two steps if the components with the highest Curie temperatures were involved. The samples from locality 12 were studied in even greater detail. Fisher's statistics (1953) were applied separately to each locality prior to and after correction for the dip of the strata. The separate samples were all assigned unit weight; assigning variable weights, indirectly proportional to the angular errors of the individual components, did not result in a significant increase of the precision parameter k.

Upper Cambrian andesite

The samples of Upper Cambrian andesite were collected from five outcrops (see Table 1, localities 1–5). The results of magnetic cleaning and multicomponent analysis for the first three outcrops are given in Krs et al. (in press).

In the first stage, the pilot samples from the first three outcrops of andesite, sampled north of Týřovice, were investigated. The group 1 samples could be partly magnetically cleaned by the alternating field, but thermal cleaning proved to be more effective. In all thermal tests, the apparent volume magnetic susceptibility κ_t of specimens heated to temperature t and cooled to room temperature, and the susceptibility of the specimen in its natural condition κ_0 were also measured. The graphs of κ_t/κ_0 versus temperature t indicate that appreciable phase changes only occurred at temperatures in excess of 600 °C, so that the mineralogical stability of the specimens in the course of thermal treatment was proved over a relatively large interval of temperatures, from 20 °C to 600 °C. The pilot group 2 samples also showed that AF demagnetization was not very effective. The group 3 samples displayed mutually different magnetic properties, and some also contained a large percentage of recent magnetization.

However, the laboratory test results proved that it was necessary to subject whole collections of samples to systematic magnetic cleaning using alternating and, especially, thermal fields. One hundred and forty cube specimens for laboratory measurements were prepared from the original collection of andesite samples from outcrops 1, 2, and 3; only 74 of these yielded Zijderveld's component diagrams which could be used in the interpretation. Figures 5–8 illustrate the stereographic projection of NRM and RM directions in the course of demagnetization in alternating and thermal fields. Table 3 gives the mean RM directions for the alternating and thermal fields at which optimum cleaning was achieved. The RM directions thus derived were interpreted as pre-Variscan (Krs et al., in press). The component diagrams indicate that most andesite samples have a two-component, occasionally four-component, RM. A number of samples proved to have a strong component of recent magnetization which could be easily removed by the thermal field. Figure 9 shows the palaeomagnetic directions of all samples studied. The mean direction is clearly not representative of the whole collection of samples, apparently because of the effect of the palaeosecular variation, or other effects. For this reason, the

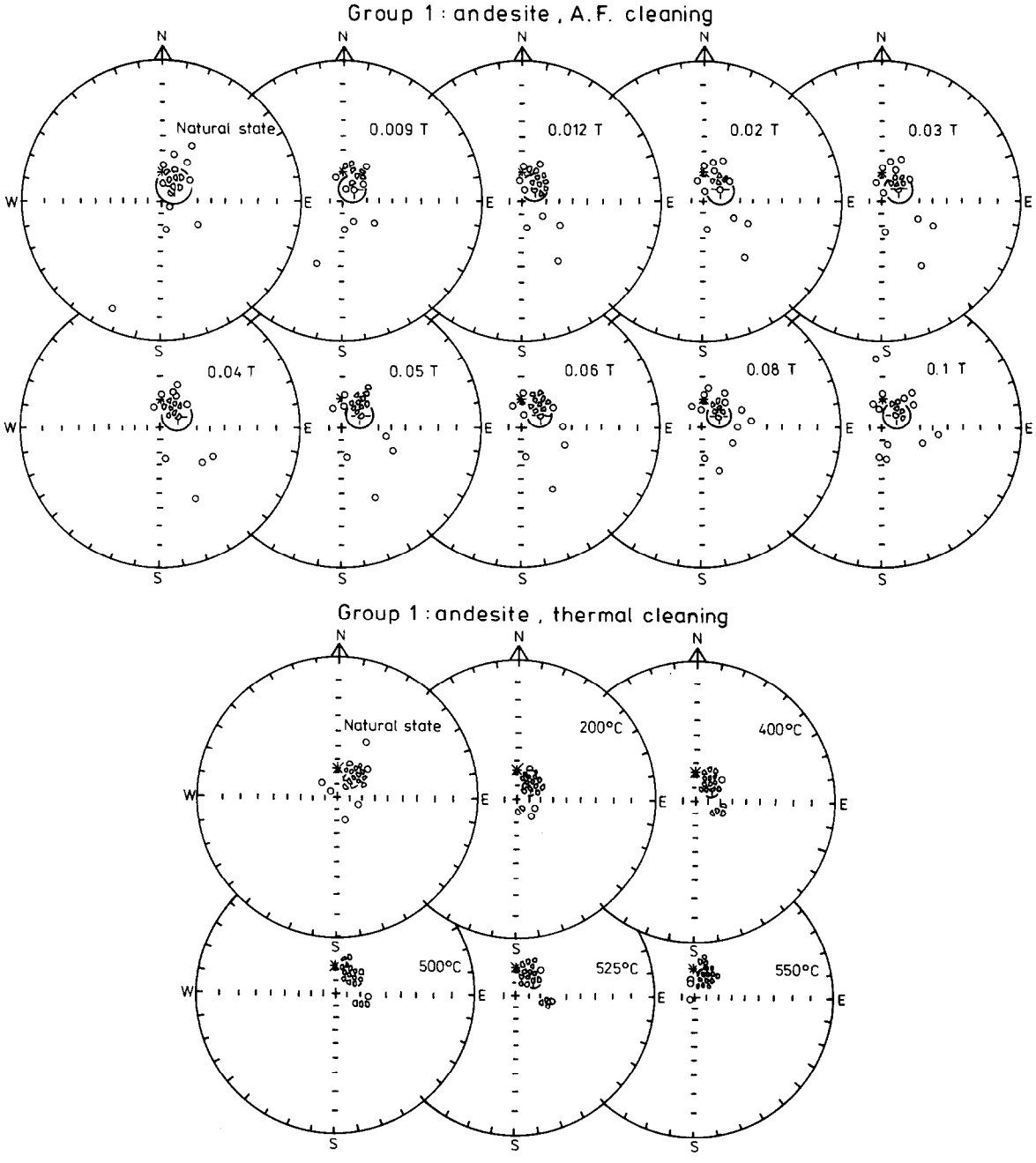


Fig. 5. AF and thermal cleaning of samples of Upper Cambrian andesite, Group 1 (cf. Table 1).

collection of andesite samples was increased in the next stage by sampling further outcrops.

Additional samples of Upper Cambrian andesite were collected from the Zaječov and Jívina localities (sample groups 4 and 5, see Table 1). Figure 10 shows the component diagrams for the pilot samples, similar diagrams being obtained for all the other samples. The andesite samples from both outcrops have a high percentage of viscous magnetization which can be removed in relatively low temperature fields of 20 °C to 200 °C. These rocks require a high magnetic vacuum for thermal cleaning, especially for reproducing the directions of the partial RM moments in higher temperature

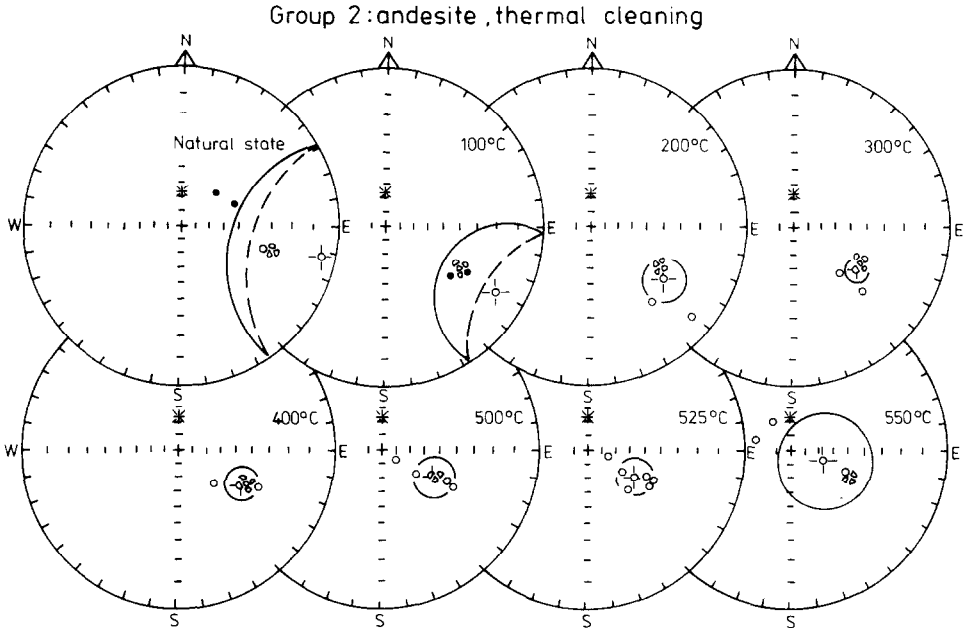


Fig. 6. Thermal cleaning of samples of Upper Cambrian andesite, Group 2 (cf. Table 1).

intervals. The MAVACS system equipped with the sensitive ROCOMA sensor provides these conditions. The temperature intervals in which the directions of the characteristic magnetization of the separate samples are derived differ slightly: for group 4 from the Zaječov locality temperature interval ranges from 400 °C to 625 °C and for group 5 from the Jívina locality from about 200 °C

TABLE 3

Palaeomagnetic directions for different groups of samples of Upper Cambrian andesite

Mean palaeomagnetic directions		α_{95} (degrees)	k	n	Notes
declination (degrees)	inclination (degrees)				
55.43	−72.91	9.60	14.79	17	Group no. 1, Fig. 5: AF cleaning at 0.08 T
22.97	−70.43	3.71	87.57	18	Group no. 1, Fig. 5: thermal cleaning at 550 °C
123.01	−37.79	6.99	92.92	6	Group no. 2, Fig. 6: thermal cleaning at 300 °C
106.01	−39.56	6.00	102.33	7	Group no. 3, Fig. 7: AF cleaning at 0.03 T
101.64	−38.74	3.52	65.79	26	Group no. 3, Figs. 7 and 8: thermal cleaning at 400 °C

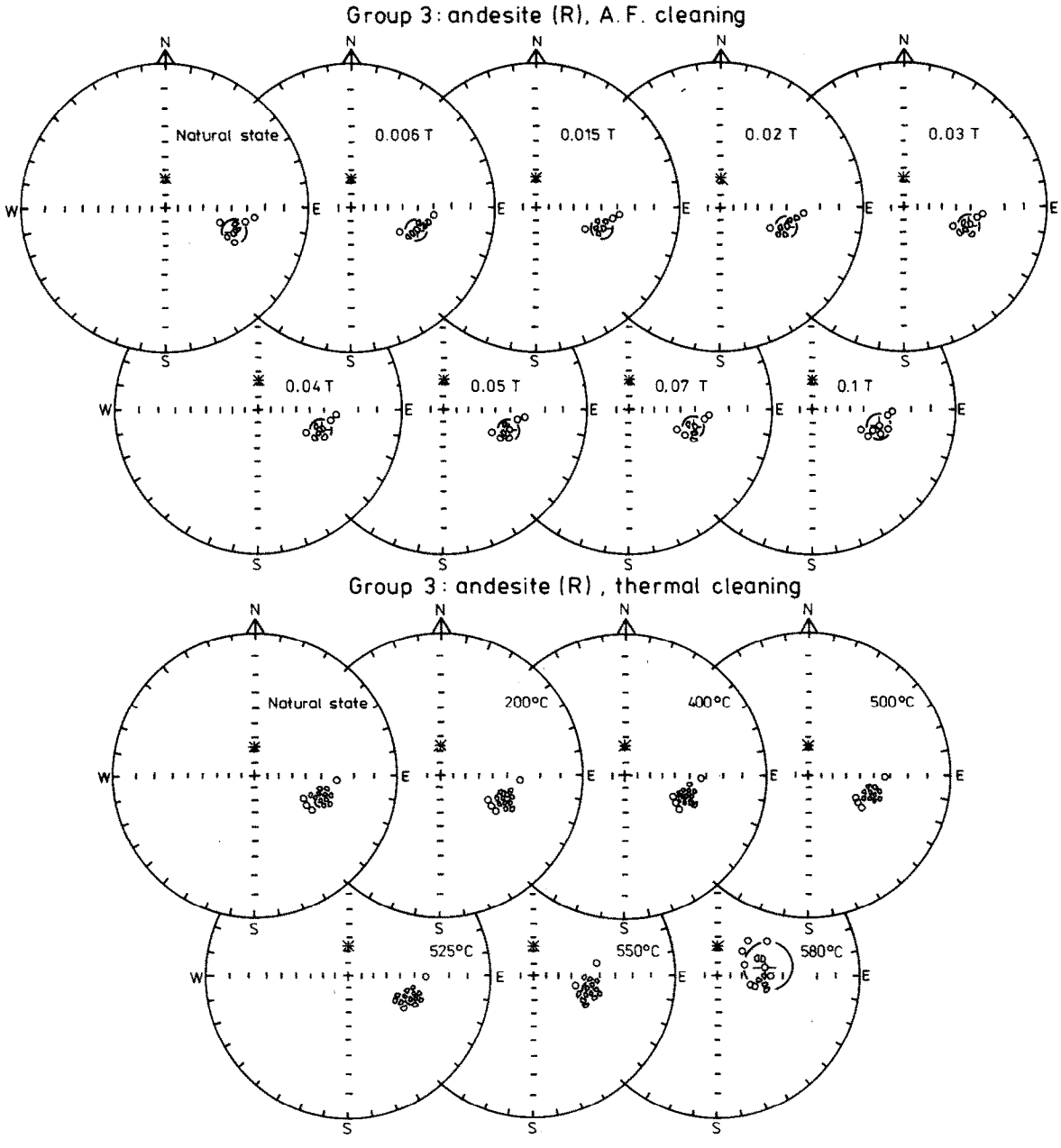


Fig. 7. AF and thermal cleaning of samples of Upper Cambrian andesite, Group 3 (cf. Table 1). NRM is reversely polarized (R).

to 525°C (see Fig. 11). However, these directions are close to the directions of the Carboniferous palaeomagnetic field in the Bohemian Massif; the directions derived in higher temperature intervals display a larger scatter and correspond to Carboniferous–Permian palaeomagnetic directions. The multicomponent analysis proved that these rocks did not preserve the older palaeomagnetic directions.

Although the palaeomagnetic directions of the andesite samples of groups 1, 2 and 3 are based on the analysis of relatively extensive collections of samples, multicomponent analysis being applied to all samples, the results represent data only from three outcrops. Such data are insufficient for deriving the mean palaeomagnetic direction in palaeovolcanics and for computing the position of the palaeomagnetic pole, because the data do not

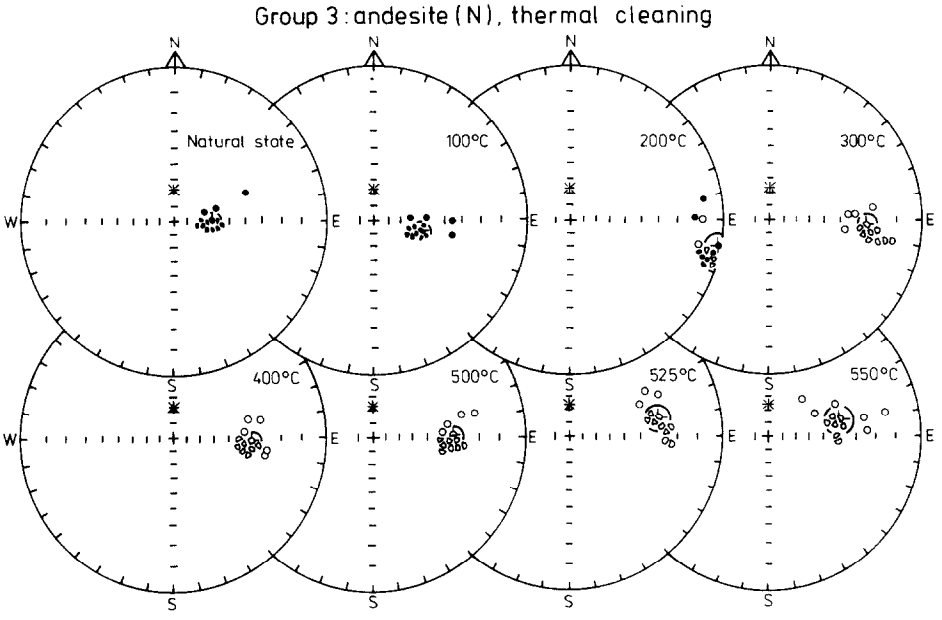


Fig. 8. Thermal cleaning of samples of Upper Cambrian andesite, Group 3 (cf. Table 1). NRM is normally polarized (N) due to strong recent components.

guarantee that the effects of the palaeosecular variation, local tectonic effects, etc., are averaged out. The results obtained from the Upper Cambrian andesite samples are, therefore, considered to be tentative, and further sample collections from other areas of the Barrandian will be added in future. Table 3 summarizes the directions of the RM of the magnetically cleaned andesite samples from the three outcrops mentioned above.

Middle Cambrian sediments

Oriented hand samples of Middle Cambrian sediments, namely greywackes, fine-grained greywackes and shales, were collected from 16 outcrops (see Table 1, localities 6–21). In an earlier study (Krs et al., in press), only the group 11

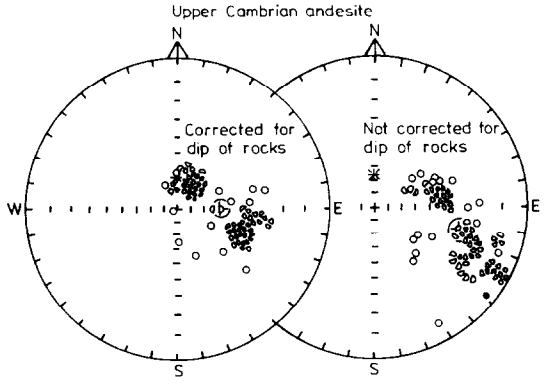


Fig. 9. Palaeomagnetic directions of the samples of Upper Cambrian andesite from localities 1, 2 and 3.

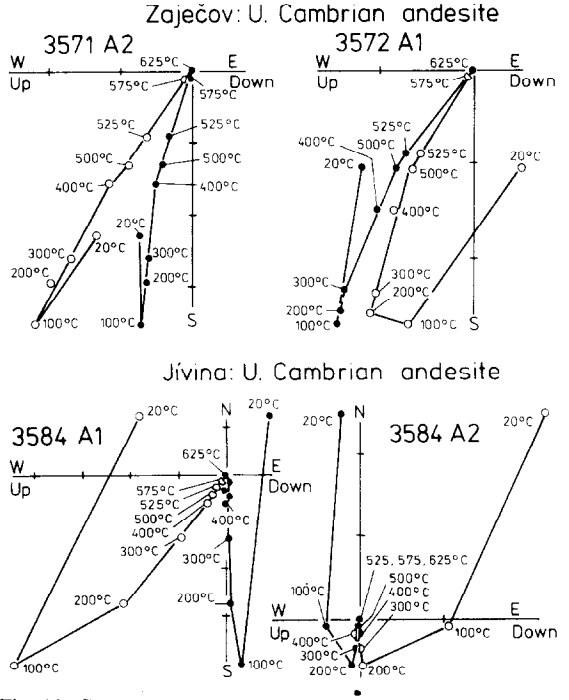


Fig. 10. Component diagrams for representative pilot samples of Upper Cambrian andesite from Zaječov and Jívina localities. These samples do not retain Cambrian palaeomagnetic directions.

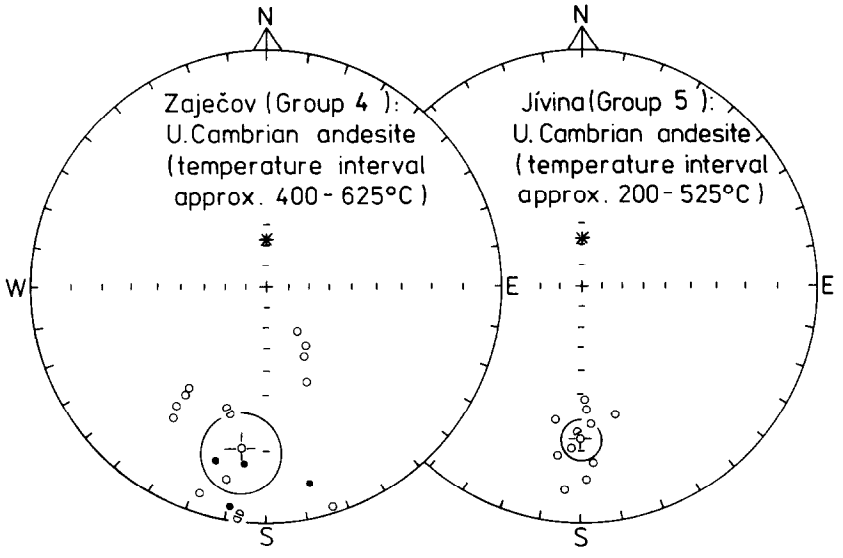


Fig. 11. Mean directions of characteristic magnetization of samples of Upper Cambrian andesite for Zaječov and Jívina localities derived by multicomponent analysis. These directions correspond to Permian and/or Permo-Carboniferous palaeomagnetic directions in the Bohemian Massif.

sediments, which could be cleaned by a thermal field, were studied (see Table 4). The subject of this paper is the study of the palaeomagnetism of the Middle Cambrian sediments from all the other outcrops. The group 12 samples were thermally cleaned in a similar way. Table 5 incidates that optimum cleaning was achieved at a temperature of 460 °C. In this group, the data obtained by the above-mentioned method were compared with the results of multicomponent analysis. Four magnetization components were found in this group (see Table 6). The first component is exceptionally soft and can be affected even by a laboratory field. The second component (mostly in the 100 °–300 °C interval) roughly corresponds to the recent geomagnetic field. Only seven samples are shown to have the third component, the directional characteristic of which is very poor. Mostly at temperatures above 460 °C, a characteristic component of magnetization was observed, the mean direction of which is almost identical with the mean direction of magnetization of the samples cleaned at 460 °C (Table 5). The mean direction of this magnetization will be compared with the results from

TABLE 4

Mean directions of remanent magnetic polarization of natural and thermally cleaned samples of Middle Cambrian greywacke, nos. 3077A-3092A, Group 11

Temperature (°C)	Mean directions		α_{95} (degrees)	k	n
	declination (degrees)	inclination (degrees)			
20	44.37	35.87	5.14	17.34	47
100	84.43	0.12	9.54	5.74	47
200	95.66	−66.72	4.45	22.83	47
300	64.83	−73.34	2.57	66.65	47
400	31.74	−72.39	2.78	56.85	47
450	26.16	−71.25	2.64	63.17	47
530	341.65	−43.77	6.53	11.12	47

TABLE 5

Mean directions of RMP of thermally cleaned samples of Middle Cambrian greywacke, nos. 3093A-3113A, Group 12

Temperature (°C)	Mean direction of remanent magnetic polarization		α_{95} (degrees)	k	n
	declination (degrees)	inclination (degrees)			
20	40.6	23.1	10.5	11.7	18
100	91.6	−36.1	26.0	2.7	18
150	88.1	−74.9	12.5	8.7	18
200	91.3	−84.2	6.4	29.8	18
250	48.6	−86.1	4.9	51.6	18
300	19.8	−86.0	3.8	85.5	18
385	7.1	−82.1	3.5	99.6	18
460	3.3	−78.7	3.1	127.0	18
520	357.5	−80.4	3.2	118.7	18

other outcrops as a palaeomagnetic direction.

With regard to all the other outcrops of sedimentary rocks, the pilot samples were studied first; rock outcrops 13, 14 and 16–21 yielded reproduceable results. By way of illustration, Fig. 12 shows the results of measuring selected pilot samples; similar results were obtained for all the other samples of the above-mentioned groups. Zijderveld's component diagrams have been plotted for each sample below the above-mentioned graphs. These graphs and diagrams clearly show the basic parameters which can be used to study the palaeomagnetism of the samples: the normalized M_t/M_0 graphs and component diagrams indicate the purity of the thermal demagnetization, and the κ_t/κ_0 graphs prove that no significant mineralogical changes took place in the magnetically active minerals in the course of the thermal treatment. Table 7 summarizes the basic directional data for the pilot samples, calculated using Fisher's statistics (1953) for the temperature intervals in which characteristic components of magnetization, useable for further interpretation, were found. The normalized graphs and component diagrams clearly indicate that many of the samples contain an exceptionally high percentage of soft components which can be removed by a relatively low temperature field of 20°–150°C. In

TABLE 6

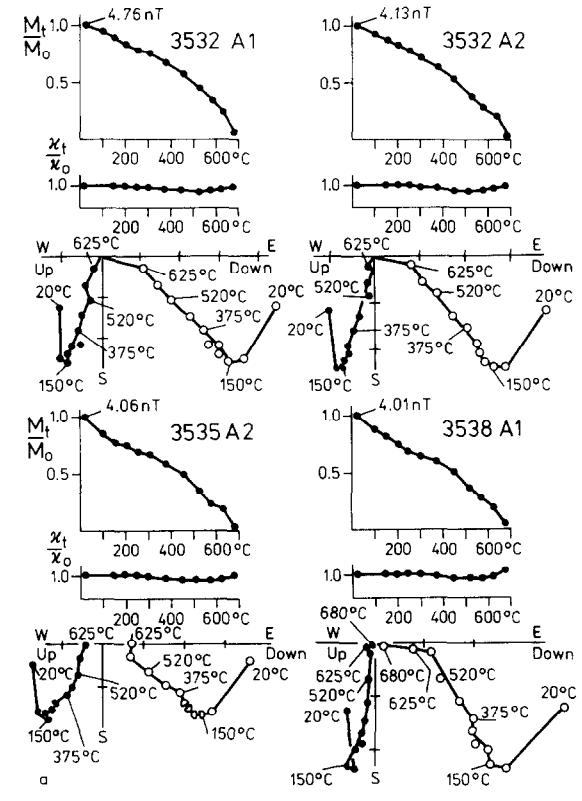
Mean directions of respective components of RMP of samples of Middle Cambrian greywacke, nos. 3093A-3113A, Group 12

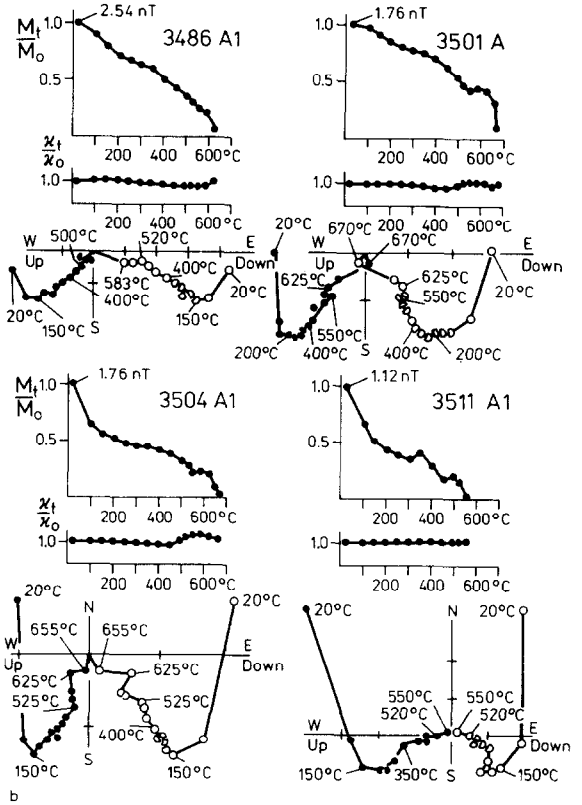
Component of RMP	Temperature interval	Mean direction of remanent magnetic polarization		α_{95} (degrees)	k	n	Correction for dip of rocks
		declination (degrees)	inclination (degrees)				
First	20°–100°C	30.1	42.8	4.1	53.8	24	corrected
		14.3	31.9	3.8	60.2	24	not corrected
Second	prevailingly	111.3	74.3	6.3	26.2	21	corrected
	100°–300°C	0.1	79.2	6.2	27.2	21	not corrected
Third	Prevailingly	166.7	−6.7	49.0	2.5	7	corrected
	300°–460°C	167.2	11.9	48.9	2.5	7	not corrected
Fourth	Prevailingly	0.4	−79.7	2.6	171.2	18	corrected
	> 460°C	109.1	−74.6	2.7	164.6	18	not corrected

TABLE 7

Palaeomagnetic directions of pilot samples of Middle Cambrian greywacke resulting from multicomponent analysis, Barrandian, cf. Table 1

Sample nos.	Group no.	Directions of partial RMP corrected for dip of rocks				Directions of partial RMP not corrected for dip of rocks				Temperature (° C) interval in which respective partial component of RMP was determined
		D (degrees)	*I* (degrees)	α (degrees)	*k*	*D* (degrees)	*I* (degrees)	α (degrees)	*k*	
3486A1, 3486A2, 3491A1, 3491A2	13	250.3	38.4	16.9	30.7	232.4	48.9	14.8	39.6	150–580
3493A1, 3493A2, 3494A1, 3494A2	14	33.4	−59.5	6.8	183.1	6.1	−44.9	6.8	183.1	520–670
3501A1, 3504A1, 3504A2, 3510A1	16	211.3	26.4	7.9	137.2	222.6	57.4	12.1	58.5	150–580
3511A, 3512A1, 3515A1, 3515A2	17	238.6	57.1	23.2	16.7	221.0	45.6	29.3	10.8	350–520
3517A2, 3521A3,	18	191.9	70.3	12.1	58.4	201.5	52.9	12.2	57.8	200–575
3522A1, 3523A1		260.6	76.9	38.0	11.6	336.6	77.6	36.9	12.2	575–680
3526A1, 3526A2 3530A2, 3530A3	19	271.4	52.1	9.5	93.8	245.6	36.3	9.8	89.6	150–520
3532A1, 3532A2,	20	198.7	42.0	11.4	65.6	219.9	59.3	11.4	65.6	160–625
3535A2, 3538A1,		230.2	75.2	16.9	30.6	306.0	67.7	16.9	30.6	625–680
3541A1, 3541A2, 3543A1, 3543A2	21	278.6	49.3	12.7	53.2	251.6	31.6	15.1	38.0	375–680





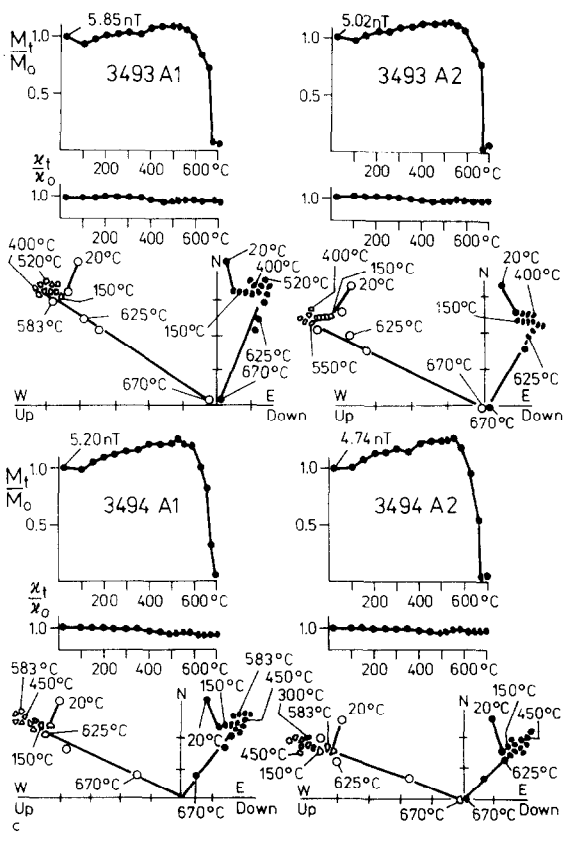


Fig. 12. Selected pilot samples for Middle Cambrian sediments. The symbol M_t represents the RM moment after thermal demagnetization at temperature t, and M_0 the NRM moment. The dependence of the magnetic susceptibility κ_t/κ_0 on temperature is denoted similarly. κ_t being the volume magnetic susceptibility after thermal demagnetization at the appropriate temperature, κ_0 being the susceptibility of the sample under natural conditions.

the next stage of the laboratory experiments, further samples were added to the pilot samples of groups 13, 14 and 16–21; the relevant results, together with the data for groups 11 and 12, are given in Table 8. Of the total number of 10 outcrops, three yielded reverse directions and seven normal directions of magnetization. The stability fold test could be applied: the rock outcrops corrected for the dip yielded $\alpha = 12.49°$, $k = 15.91$, and the rock outcrops not corrected for the dip yielded $\alpha = 15.48°$, $k = 10.69$. This calculation was carried out for the mean directions of 10 outcrops ($N = 10$, see Table 9). Figure 13 shows the mean palaeomagnetic directions for the dis-

TABLE 8

Mean palaeomagnetic directions (locality means) of samples of Middle Cambrian greywacke, Barrandian, cf. Table 1

Group no. of samples	Mean palaeomagnetic directions		α_{95} (degrees)	k	n	Notes
	declination (degrees)	inclination (degrees)				
11	64.8	−73.3	2.6	66.7	47	thermal cleaning
12	0.4	−79.7	2.6	171.2	18	thermal cleaning: multicomponent analysis
13	247.2	37.3	16.6	9.4	10	
14	75.3	−53.9	14.4	12.3	10	
16	218.6	28.8	8.5	33.3	10	
17	254.8	60.4	11.7	23.4	8	
18	210.6	71.2	8.9	30.3	10	
19	252.5	52.2	12.0	17.1	10	
20	204.3	44.0	5.9	67.2	10	
21	244.0	67.0	10.3	23.1	10	

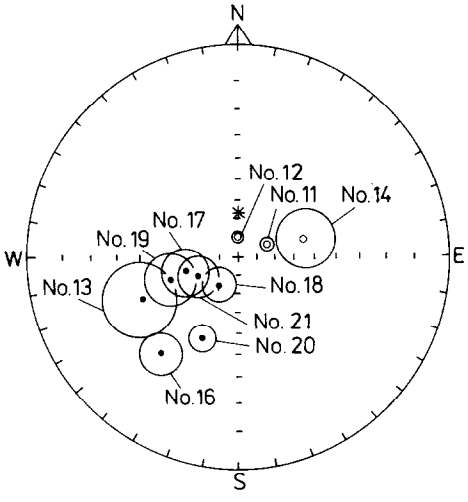


Fig. 13. Mean palaeomagnetic directions and circles of confidence at the 95% probability level (Fisher, 1953) for ten outcrops of Middle Cambrian sediments (see Tables 1 and 8).

cussed outcrops of the Middle Cambrian sedimentary rocks of the Barrandian.

Main results and discussion

Table 9 summarizes the palaeomagnetic data. The palaeomagnetic directions of the Upper Cambrian andesites were derived from an extensive set of the samples which were collected from three outcrops of one large andesite extrusion. Geologically, one lava flow was assumed, but the palaeomagnetic data indicate that two lava flows may have been involved (see Fig. 9). Consequently, it is necessary to consider the position of the palaeomagnetic pole, determined for the Upper Cambrian, as tentative and the collections of samples will have to be increased by adding samples from further outcrops in other areas of the Barrandian. Two further andesite outcrops in the Zaječov and Jívina localities yielded samples with an exceptionally high percentage of magnetically soft components. The multicomponent analysis showed that the direction of the characteristic magnetization corresponds to the Carboniferous or Permo-Carboniferous palaeomagnetic direction in the Bohemian Massif, and that the original Cambrian palaeomagnetic directions were not preserved in these rocks.

The samples of the Middle Cambrian greywackes and shales from 10 outcrops yielded reproduceable results (see Tables 1 and 8). Whole collections of samples were subjected to thermal cleaning using a high magnetic vacuum thermal demagnetizer. The RM was subjected to multicomponent analysis: most samples displayed a number of magnetization components, usually four (see, e.g., Table 6).

The palaeomagnetic directions and the position of the palaeomagnetic pole, calculated from the mean directions at the localities ($N = 10$), are considered to be representative of the Middle Cambrian rocks of the Barrandian. Figure 1 shows the palaeogeographic latitudes, the mean palaeo-

TABLE 9

Mean palaeomagnetic directions and pole positions for Upper Cambrian andesite (preliminary) and Middle Cambrian sediments determined in this study, Barrandian, Bohemian Massif *

Age, lithology	Geographic coordinates		Palaeomagnetic directions		α_{95} (degrees)	k	n	N	Palaeomagnetic pole positions		Ovals of confidence		Notes, stability tests
	latitude	longitude	declination (degrees)	inclination (degrees)					palaeolatitude	palaeolongitude	δp (degrees)	δm (degrees)	
Upper Cambrian andesite	49.983° N	13.783° E	89.88	−58.16	5.77	9.16	*74*	3	28.64° S	131.23° E	6.29	8.52	Pole position calculated from palaeomagnetic directions of samples (n = 74); AF and thermal cleaning, multicomponent analysis of pilot samples
Middle Cambrian greywacke	49.943° N	13.747° E	54.31	−58.67	12.49	15.91	143	*10*	11.29° S	153.96° E	13.79	18.56	Pole position calculated from locality means (N = 10); thermal cleaning and multicomponent analysis of all samples
	49.943° N	13.747° E	54.37	−64.34	3.47	12.56	*143*	10	16.99° S	157.67° E	4.44	5.55	Pole position calculated from palaeomagnetic directions of samples (n = 143); thermal cleaning and multicomponent analysis of all samples

* n = number of samples; N = number of localities.

magnetic field being assumed to have had a dipolar character. With the accuracy given in Table 9, these palaeogeographic latitudes roughly agree with the palaeozoogeographic zones. A cold-water environment is characteristic for the Bohemian fauna, but arctic conditions may be ruled out due to the amount of primary red sediments, mostly of continental origin.

The data presented in this paper should be discussed in relation to Cambrian palaeomagnetic results which refer to Fennoscandian rocks and the Armorican Massif. Bylund (in press) reviews the palaeomagnetic results obtained so far from investigations of Fennoscandian Cambrian rocks. The data obtained on Nexö sandstone, Fen and Alnö carbonatite complexes are widely scattered and, with the additional data from Norway, only a tentative apparent polar wander path could be suggested. These data, as compared to those from Armorica (Perroud, 1985), England and Wales, generally indicate large movements of the respective geological blocks, and may be interpreted as the result of the break-up of a Late Proterozoic supercontinent at the beginning of the Cambrian (Piper, 1985; cf. Bylund, in press).

Figure 14 shows the palaeolatitudes in dependence on geological age, calculated from the positions of palaeomagnetic poles for the Barrandian area. The palaeolatitudes marked 1–5 were derived from the palaeomagnetism of the rocks of the Bohemian Massif and are all substantiated by abundant statistical data. These data indicate a more or less smooth movement of the Barrandian block from about 40 °S, beginning in the Middle Cambrian and continuing to the present. However, the existing values should be supplemented with the palaeomagnetism relating to the Silurian, Devonian and Lower Cambrian. The palaeomagnetic research into these formations of the Barrandian is being continued, and research into the palaeomagnetism of the Proterozoic rocks of the Barrandian will be undertaken in order to clarify the palaeogeographic position of the bordering continental blocks of the proto-Atlantic Ocean. However, the existing Barrandian data already display a certain affinity to those of the Armorican Massif (cf. Perroud, 1985), the palaeolatitudes referring to the Barrandian Cambrian being somewhat lower than those derived for the Armorican Massif.

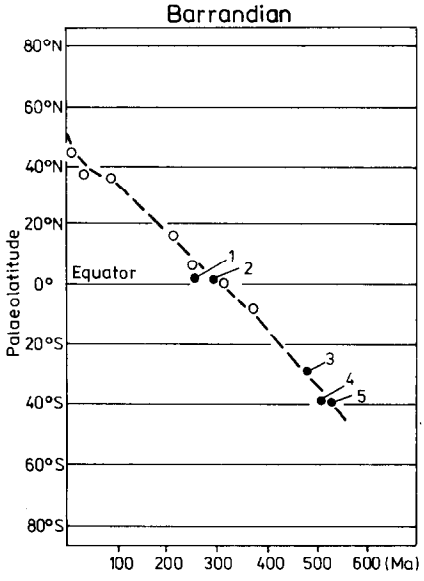


Fig. 14. Palaeolatitudes of the Barrandian. Open circles indicate palaeolatitudes for the Barrandian, but calculated from statistically derived pole positions of the North European Platform (Krs, 1982). Full circles indicate palaeolatitudes derived on rocks of the Bohemian Massif: *1*—L. Permian; *2*—U. Carboniferous (Krs, 1968); *3*—L. Ordovician of the Barrandian; *4*—U. Cambrian (preliminary) of the Barrandian; *5*—M. Cambrian of the Barrandian.

The authors of this paper have put emphasis on the multicomponent analysis of the NRM, and have used a thermal demagnetizer with a high magnetic vacuum to render the reliability of the palaeomagnetic parameters derived as high as possible. In the course of the next few years, it will become possible to correlate the palaeomagnetic data of the Barrandian, Armorica and Baltica with the purpose of producing a palaeotectonic model of the pre-Variscan Laurasian formations.

References

Bylund, G., in press. Palaeomagnetism of Fennoscandian Palaeozoic rocks. Geol. Fören. Stockholm Förh.

Chlupáč, I. and Havlíček, V., 1965. Kodymirus n.g., a new aglaspid merostone of the Cambrian of Bohemia. Sb. Geol. Věd, Paleontol., 6: 7–20.

Fisher, R., 1953. Dispersion on a sphere. Proc. R. Soc., Ser. A, 217: 295–305.

Hagstrum, J.T., Van der Voo, R., Auvrey, B. and Bonhommet, N., 1980. Eocambrian–Cambrian palaeomagnetism of the

Armorican Massif, France. Geophys. J. R. Astron. Soc., 61: 489–517.

Halousek, J., Pešina, B. and Příhoda, K., 1985. A new rotating coil magnetometer. 5th General Assembly of IAGA/IAMAP, Prague, 2: 351 (abstr.).

Havlíček, V., 1971. Stratigraphy of the Cambrian of Central Bohemia. Sb. Geol. Věd, Geol., 20: 7–52.

Havlíček, V., 1980. Development of Paleozoic basins in the Bohemian Massif (Cambrian–Lower Carboniferous). Sb. Geol. Věd, Geol., 34: 31–65.

Jelínek, V., 1966. A high sensitivity spinner magnetometer. Stud. Geophys. Geod., 10: 58–78.

Jelínek, V., 1973. Precision A.C. bridge set for measuring magnetic susceptibility and its anisotropy. Stud. Geophys. Geod., 17: 36–48.

Kent, J.T., Briden, J.C. and Mardia, K.V., 1983. Linear and planar structure in ordered multivariate data as applied to progressive demagnetization of palaeomagnetic remanence. Geophys. J. R. Astron. Soc., 1983: 593–621.

Krs, M., 1968. Rheological aspects of palaeomagnetism? Int. Geol. Congr., 13th, Prague, 5: 87–96.

Krs, M., 1969. Paleomagnetismus. Knih. Ústřed. Ústavu Geol., 40: 207 pp.

Krs, M., 1978. Some general aspects of palaeomegnetic stability tests carried out on Cambrian rocks (Bohemian Massif). Electromagnetic Field of the Earth. Proc. Symp. Slovak Acad. Sci., Bratislava, 69–88.

Krs. M., 1982. Implication of statistical evaluation of Phanerozoic palaeomagnetic data (Eurasia, Africa). Řada MPV, 92, 3, Academia, Praha, 86 pp.

Krs, M., Krsová, M., Pruner, P. and Havlíček, V., in press. Palaeomagnetism, magnetism and palaeogeography of the Upper Cambrian rocks of the Barrandian area, Bohemian Massif. Sb. Geol. Věd, Užitá Geofyz.

Kukal, Z., 1971. Sedimentology of Cambrian deposits of the Barrandian area (Central Bohemia). Sb. Geol. Věd, Geol., 20: 53–100.

Perroud, H., 1985. Synthése des résultats paléomagnétique sur le Massif Armoricain. Hercynica, 1: 65–72.

Piper, J.D.A., 1985. Continental movements and breakup in Late Precambrian–Cambrian times: prelude to Caledonian orogenesis. In Gee, D.G. and Sturt, B.A. (Editors), The Caledonide Orogen—Scandinavia and Related Areas. Willey, New York, pp. 19–34.

Waldhausrová, I., 1966. The volcanites of the Křivoklát–Rokycany zone. In: Paleovolcanites of the Bohemian Massif. Proc. Přírodověd. Fak. Univ. Karl., Praha, 145–151.

Waldhausrová, I., 1972. The chemistry of the Cambrian volcanics in the Barrandian area. Krystalinikum, Praha, 8: 45–75.